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Working paper.

Motivation

The 45Z Puzzle

Two numbers from the same SE-US supply chain

- **Observed** 45Z producer credit (IRA 2022, GREET 2024 default CI):
 $\sigma_{45Z}^{\text{obs}} \in [\$1.30, \$1.45]/\text{gal}.$
- **Welfare-equivalence** subsidy (Paper 1 planner CVaR-optimum):
 $\sigma_{\text{planner}}^* = \lambda_{\text{planner}} \cdot \pi_H \cdot (a_H - a_L) = \$0.52/\text{gal}.$

Ratio $\approx 2.5\times$. **Gap** $\approx \$0.84/\text{gal}.$

Why is the credit so much larger than what welfare economics says?

Source: Li and Yu (2026) for $\sigma_{\text{planner}}^*$; IRS Notice 2024-49 for $\sigma_{45Z}^{\text{obs}}$.

The Policy Window is Open — And Narrow

Three legislative moments

- **IRA 2022:** 45Z enacted at \$1.75/gal (linear per-gallon, CI-sliding scale).
- **H.R. 1 (One Big Beautiful Bill Act), July 2025:** cut from \$1.75 to \$1.00/gal.
- **Q3 2026 CFR rulemaking cycle:** **currently OPEN**. The rulemaking record has not addressed: *linear per-gallon vs. state-contingent transfer form*.

The window for this paper

The H.R. 1 cut did not merely trim rent; at \$1.00/gal a pioneer-plant SAF facility no longer clears its hurdle IRR at the median SAF price projection. The CFR cycle is the chance to fix *level* or *form* — but to do either, we need to know what the level is paying for.

Two Research Questions

- 1 **Decomposition:** What does the observed \$1.36/gal credit pay for?
 - How much is welfare-equivalent transfer?
 - How much is risk premium?
 - How much is project-finance hurdle premium?
- 2 **Form:** Is the linear per-gallon form welfare-optimal under empirically calibrated producer risk aversion and project finance?
 - Or does a Contract-for-Difference (CfD) with break-even floor dominate?

This talk answers both, in closed form, with an empirical calibration, and with a concrete CFR rulemaking proposal.

Roadmap

- 1 **Positioning:** Cullen 2013, Wu 2025, Borch–Wilson, Holmstrom–Milgrom.
- 2 **Model:** Stackelberg + form-commitment + mean-CVaR.
- 3 **Main Result 1:** Closed-form σ_{prod} and 12.2% / 87.8% wedge attribution.
- 4 **Main Result 2:** CfD-with-floor weak Pareto dominance (strict at $\text{MCPF} > 1$).
- 5 **Robustness:** T , r_h , MCPF, depreciation convention, pathway heterogeneity, asymmetric info.
- 6 **Policy Product:** \$3–\$9/ton CO₂ welfare gain, Title XVII 46% fiscal substitution, IRC §45Z(g) amendment.

Positioning

The Closest Methodological Anchor: Cullen (2013)

Cullen 2013 (AEJ:EP) on the wind PTC

- Same template: *financial-equivalence margin* (does the credit clear the project's hurdle IRR?), empirically calibrated.
- Welfare evaluated at a **risk-neutral** project-finance margin.
- Cullen explicitly defers *producer risk aversion* to future work.

What this paper fills in

- Same template, applied to 45Z (SAF).
- Risk aversion accounts for **12.2%** of the wedge at $T = 10$.
- The dominant term — **87.8%** at $T = 10$ — is a project-finance premium that Cullen's risk-neutral framework would miss entirely.
- Decomposition would be qualitatively incorrect (not just imprecise) without risk aversion.

Direct Predecessor: Wu et al. (2025)

Wu et al. (NBER w34326, Oct 2025): *level* question

- Detailed model of U.S. fuel markets + aviation demand.
- Result: per-gallon producer credit is among the *three least-cost instruments* to hit the 3 B gal/yr SAF Grand Challenge target.
- Cost-per-ton-of-abatement comparison across a menu of designs.

What Wu et al. do not address: the *form* question

- Wu et al. take instrument design as exogenous: linear per-gallon.
- This paper endogenizes the σ -form choice in Stage 0b.
- **The two papers are complementary: Wu et al. determine the level; this paper determines the form.**

Theoretical Anchors: Risk-Sharing and Its Failure Mode

Affirmative side

- Borch (1962): Pareto-optimal risk-sharing equalizes the marginal-utility ratio across states.
- Wilson (1968): extends to syndicates with heterogeneous beliefs / utilities.
- Newbery (2023): CfD welfare dominance in renewable-electricity auctions (different mechanism).

Failure-mode benchmark

- Holmstrom and Milgrom (1987) (H-M)
Theorem 7: under *CARA-Normal preferences* and an *unrestricted contract space*, the optimal contract is **linear**.
- Two H-M conditions break in our setting:
 - (a) mean-CVaR (not CARA-Normal);
 - (b) restricted policy class $\mathcal{S}$ (IRC §45 statutory).
- Dropping these restores linear optimality (H-M escape route).

Borch–Wilson buys the proof; H-M tells us exactly which structural assumptions are load-bearing.

Positioning Map

	Cullen 2013	Wu 2025	This paper
Credit	wind PTC	45Z (SAF)	45Z (SAF)
Question	<i>level</i>	<i>level</i>	<i>decomposition + form</i>
Risk aversion	deferred	no	yes (mean-CVaR)
Project finance	yes (risk-neutral)	no	yes (hurdle IRR r_h)
Welfare functional	expected	expected	mean-CVaR
Instrument class	linear (fixed)	linear (fixed)	restricted $\mathcal{S}$ (4 forms)
Closed form?	partial	no	yes (12.2% / 87.8% wedge)
Form-dominance theorem?	no	no	yes (Borch–Wilson, $\mathcal{S}$)

Inheritance: $\sigma_{\text{planner}}^* = \$0.52/\text{gal}$ from Li and Yu (2026); same SE-US 248/34 supply chain.

Model

Timing: Stackelberg with Form-Commitment

Four-stage game (regulator leads)

- 1 **Stage 0a** — Regulator publicly observes producer financial primitives $(r_h, e, r_e, r_d, \tau, \kappa_{\text{CAPEX}})$. *No choice.*
Justification: Assumption A10 (public info). 80 harmonized TEAs + Lazard 2024 + NREL ATB 2024.
- 2 **Stage 0b** — Regulator irrevocably commits a *form* $\sigma(\cdot) \in \mathcal{S}$ (4-element design space).
A9: form-commitment, not level-commitment. H.R. 1 revised level, preserved form.
- 3 **Stage 1** — Producer invests $y \in \{0, 1\}^{|D|+|R|}$ (SE-US 248 depots / 34 refineries) subject to $\text{NPV} \geq 0$ at hurdle r_h (A13).
- 4 **Stage 2** — Nature draws $\omega \sim F$; producer chooses $Q(\omega)$.

Backward induction: $Q^* \rightarrow y^* \rightarrow \sigma^*$. Subgame-perfect Stackelberg equilibrium.

The Restricted Policy Class $\mathcal{S}$ (Assumption A11)

- 1 **Linear:** $\sigma^{\text{lin}}(\omega) = \sigma_0$
Observed 45Z form.
- 2 **CfD-strike:** $\sigma^{\text{strike}}(\omega; K) = \max\{K - p_{\text{SAF}}(\omega), 0\}$
- 3 **CfD-with-floor:** $\sigma^{\text{floor}}(\omega; K, s_0) = \max\{K - p_{\text{SAF}}(\omega), 0\} + s_0$
UK Contracts for Difference; Newbery (2023).
- 4 **Capacity:** $\sigma^{\text{cap}}(\omega) = \kappa \cdot y$

Why restrict $\mathcal{S}$? (Statutory, not modeling crutch)

- IRC §45 prescribes *per-unit-output* architecture (kWh for 45Y, kg for 45V, ton CO₂ for 45Q, gallon for 45Z).
- Two-part tariffs excluded from §45 (administered separately under DOE Title XVII).
- Capacity-contingent under §45 technically allowed but no IRA-era precedent.
- Dropping A11 + moving to CARA-Normal restores H-M linear optimality.

Payoffs

Producer mean-CVaR objective:

$$\text{Obj}_{\text{prod}}^{\text{op}}(y, \sigma) = (1 - \lambda_{\text{prod}}) \mathbb{E}[\pi^{\text{op}}] + \lambda_{\text{prod}} \text{CVaR}_{\alpha}[\pi^{\text{op}}]$$

Regulator mean-CVaR welfare:

$$\max_{\sigma \in S} (1 - \lambda_{\text{planner}}) \mathbb{E}[W] + \lambda_{\text{planner}} \text{CVaR}_{\alpha}[W], \quad W = \text{CS} + \text{PS} + \text{ENV} - \mu_W \cdot T(\sigma, Q)$$

Producer project-finance participation (A13, the key structural addition):

$$\text{NPV}(\sigma; r_h, \kappa_{\text{CAPEX}}, u, T) = \text{AF}(r_h, T) \mathbb{E}_{\text{prod}} \left[\frac{\text{CF}(\sigma, Q^*; \omega)}{\bar{Q}} \right] - \frac{\kappa_{\text{CAPEX}}}{u} \geq 0.$$

σ_{prod} is the minimum constant σ_0 that makes this = 0.

Two-State Collapse (for closed-form propositions)

20-bin KDE collapsed to $\{H, L\}$ at $\alpha \geq \pi_H$

$$\omega \in \{H, L\}, \quad \pi_H = 0.472, \quad \pi_L = 0.528,$$
$$a_H = \$7.16/\text{gal}, \quad a_L = \$4.97/\text{gal}, \quad p_{\text{SAF}}^- = \$6.00/\text{gal}.$$

- Justified by Li and Yu (2026) Theorem 1 calibration device.
- In the regime $\alpha \geq \pi_H$, mean-CVaR collapses to distorted-EU (**Lemma B.0** in the appendix; backup slide).
- This collapse is the technical key that makes Borch–Wilson directly applicable.

Assumption Summary

Paper 1 inheritance (A1–A8)

A1 Two-state collapse	$\omega \in \{H, L\}$ at $\alpha \geq \pi_H$.
A2 Linear inverse demand	$P^d(Q; \omega) = a(\omega) - bQ$.
A3 Convex capacity cost	$C(Q) = (k/2)Q^2$.
A4–A8	State ordering, positive net premium, CVaR confidence, bounded $\lambda_{\text{prod}}, \lambda_{\text{planner}}$, partial eq.

Paper 2 additions (A9–A13)

A9 Form-commitment	Form irrevocable; level revisable (H.R. 1 pattern).
A10 Public info	$(\lambda_{\text{prod}}, r_h, \kappa_{\text{CAPEX}})$ common knowledge.
A11 Restricted $\mathcal{S}$	4-element class; IRC §45 statutory envelope.
A12 MCPF = 1 baseline	Robustness to Dahlby [1.10, 1.30] in §6.
A13 Project-finance NPV ≥ 0	At hurdle IRR $r_h = 0.125$, $\kappa_{\text{CAPEX}} = \$11/\text{gal}$.

Main Result 1: Closed-Form σ_{prod}

Proposition: Closed-Form σ_{prod}

Proposition (Financial-equivalence subsidy via NPV = 0):

Under A1–A13 in the regime $\alpha \geq \pi_H$,

$$\sigma_{\text{prod}} = \underbrace{\frac{\kappa_{\text{CAPEX}}}{u \cdot \text{AF}(r_h, T) \cdot (1 - \tau)}}_{\text{Term 1: hurdle-IRR premium}} + \underbrace{\lambda_{\text{prod}} \cdot \pi_H \cdot (a_H - a_L)}_{\text{Term 2: risk-aversion premium}} - \underbrace{[p_{\text{SAF}}^- - c_{\text{op}}^-]}_{\text{Term 3: operating-margin offset}}$$

- **Term 1:** per-gallon CAPEX-recovery requirement at hurdle IRR. Absent from Cullen 2013, absent from Paper 1.
- **Term 2:** downside premium under producer mean-CVaR.
- **Term 3:** expected operating margin earned even without subsidy (offset).

Proof: project-finance NPV = 0 condition at hurdle IRR r_h ; closed form via linearity of Q^* in σ . *Full proof in backup.*

Calibration: Three-Source Cross-Walk

Three-source identification of $r_h = 0.125$

- **Primary anchor:** DOE SAF Grand Challenge Implementation Plan (2022) §5 — pioneer-plant equity premium for FOAK SAF.
- **Quantitative anchors:** Lazard LCOE+ 2024 after-tax WACC benchmarks (7.7–9.1%; power/H₂, carried as economy-wide clean-energy benchmarks) + Graham–Harvey hurdle-over-WACC premium; NREL ATB 2024 capital-charge factor.
- **Honest caveat:** no agency document publishes a numeric SAF hurdle rate — FAA 2021 and IATA 2023 discuss financing risk qualitatively only.
- **Envelope:** $r_h \in [0.10, 0.15]$, midpoint = 0.125.

Other primitives

- $\kappa_{\text{CAPEX}} = \$11/\text{gal}$ annual capacity ($\approx 1.2\times$ FOAK premium over the \$9.31/gal-yr n th-plant NREL anchor, Tan et al. 2015; FT-FR pathway).
- $u = 0.85$ (7440 operating hours/yr); $\tau = 0.21$ (federal corporate rate).
- $T = 10$ years (Paper 1 horizon, DOE Title XVII norm, IRS Asset Class 49.5).
- $\lambda_{\text{prod}} = 0.60$ by revealed preference: inverting the NPV-binding identity at the observed 45Z band conditional on r_h (band image [0.54, 0.69]; Farooq et al. 2025 anchors TEA primitives, not the CVaR weight).

Headline Number: $\sigma_{\text{prod}} \approx \$1.36/\text{gal}$

Substituting baseline calibration into the closed form:

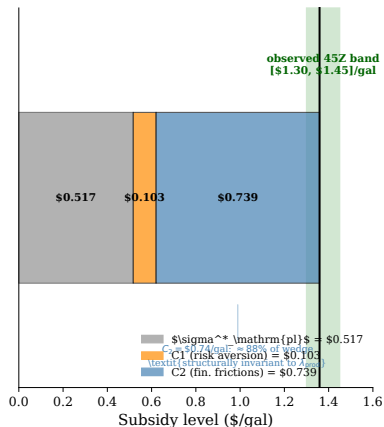
$$\sigma_{\text{prod}} = \underbrace{\frac{11}{0.85 \cdot 5.5364 \cdot 0.79}}_{=\$2.96/\text{gal}} + \underbrace{0.60 \cdot 0.472 \cdot 2.19}_{=\$0.620/\text{gal}} - \underbrace{(6.00 - 3.78)}_{=\$2.22/\text{gal}} = \boxed{\$1.36/\text{gal}}$$

Match to the empirical 45Z band

- Observed: [\$1.30, \$1.45]/gal at the GREET 2024 default CI case.
- Model: \$1.36/gal at FT-FR baseline, $T = 10$.
- Match within 5% at the point estimate. **Inside the band.**

The closed form, calibrated independently, hits the observed 45Z within 5%.

The Wedge Decomposition



$$\sigma_{\text{prod}} - \sigma_{\text{planner}}^* = \underbrace{C_1}_{\text{risk}} + \underbrace{C_2}_{\text{fin. fric.}}$$

$$C_1 = (\lambda_{\text{prod}} - \lambda_{\text{planner}}) \pi_H (a_H - a_L)$$

$$C_2 = \Delta \text{IRR}_{\text{prem}}$$

At $T = 10$:

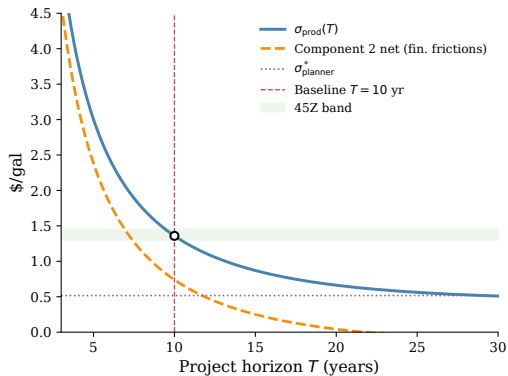
- $C_1 = \$0.103/\text{gal} \rightarrow 12.2\% \text{ risk}$
- $C_2 = \$0.739/\text{gal} \rightarrow 87.8\% \text{ fin.}$
- Wedge = $\$0.842/\text{gal}$

Dominant term: project-finance hurdle premium — not behavioral risk aversion.

Attribution Shifts Strongly with Horizon T

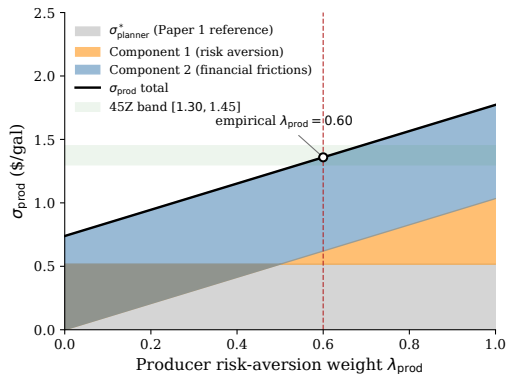
T (yr)	$AF(r_h, T)$	σ_{prod} (\$/gal)	C_1 (\$/gal)	C_2 (\$/gal)	RA share	FF share	H.R. 1 gap (\$1.00 $-\sigma_{\text{prod}}$)
5	3.5606	3.001	0.103	2.381	4.2%	95.8%	-2.00
10	5.5364	1.359	0.103	0.739	12.2%	87.8%	-0.36
15	6.6329	0.870	0.103	0.250	29.3%	70.7%	+0.13
20	7.2414	0.662	0.103	0.042	71.0%	29.0%	+0.34

- C_1 is invariant to T ; C_2 declines strictly as $AF \rightarrow 1/r_h = 8$.
- **Crossover region $T \in [15, 20]$** : attribution transitions C2-dominant $\rightarrow$ C1-dominant.
- DOE Title XVII tenor (10 yr) and FOAK project-life (10–15 yr) span the crossover.
- The 12.2%/87.8% split is a **point estimate at $T = 10$** , not a robust band.

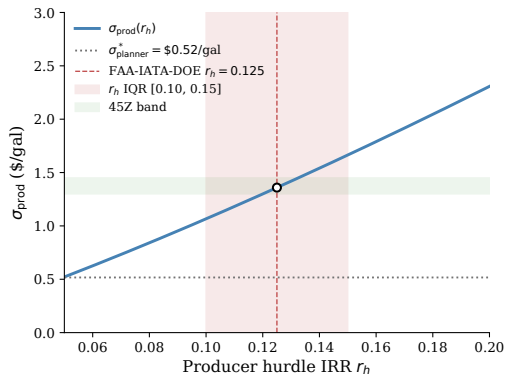
σ_{prod} vs. Horizon T 

Strictly monotone decreasing; attenuating slope; asymptote > 0 . Baseline $T = 10$ vertical dashed line.

σ_{prod} vs. Producer Risk Aversion λ_{prod}



Slope $\pi_H(a_H - a_L) = \$1.03/\text{gal}$ per unit λ_{prod} . Band image $[0.54, 0.69] \Rightarrow \sigma_{\text{prod}} \in [\$1.30, \$1.45]$ — returns the band by construction; the content is the slope.

σ_{prod} vs. Hurdle IRR r_h 

Calibrated $r_h \in [0.10, 0.15] \Rightarrow \sigma_{\text{prod}} \in [\$1.11, \$1.62]$. Observed 45Z band $[\$1.30, \$1.45]$ pins $r_h \in [0.116, 0.135]$ — subset of calibrated range.

Pathway Heterogeneity (§6.7, R&R Round 2 R1 CI3)

The 45Z band reflects CI heterogeneity, not noise

Pathway	κ_{CAPEX} (\$/gal)	$\bar{c}_{op}$ (\$/gal)	σ_{prod} (\$/gal)	45Z band status
FT-FR (SE-US forest residue)	11.0	3.78	1.36	within (Paper 1 baseline)
FT (biomass Fischer-Tropsch)	≈ 15	≈ 4.00	2.66	above (high CAPEX)
ATJ-corn (Alcohol-to-Jet)	≈ 9	≈ 3.50	0.54	below (low CAPEX)
PtL (Power-to-Liquid, ren. $\text{H}_2 + \text{CO}_2$)	≈ 25	≈ 5.00	6.35	far above (FOAK)

- **\$1.36/gal is the FT-FR pathway-specific match**, not a generic band match.
- 45Z is a CI-sliding-scale credit (IRS Notice 2024-49): high-CI-reduction pathways (PtL, FT) receive proportionally higher per-gallon credit.
- High-CAPEX pathways (PtL, FT) benefit *more* from Title XVII r_h compression (§6 robustness, conclusion).

Robustness: Depreciation Convention (§6.6)

Two conventions, one honest disclosure

- **Absorption convention** (Brandao 2025; Farooq 2025; Lazard 2024; Cullen 2013):

τD_t absorbed into κ_{CAPEX} at calibration.

$\Rightarrow \sigma_{\text{prod}} = \$1.36/\text{gal}$ (**within** 45Z band).

- **Rigorous depreciation derivation:** tax shield carried through annuity factor:

$$\sigma_{\text{prod}}^{\text{rig}} = \frac{\kappa_{\text{CAPEX}}}{u \text{AF}(1-\tau)} \left[1 - \frac{\text{AF} \tau}{T} \right] + \lambda_{\text{prod}} \pi_H (a_H - a_L) - (\bar{p} - \bar{c}_{op}) = \$1.02/\text{gal}.$$

- Absorption defended on 3 grounds: standard in TEA literature, matches MACRS 7-yr front-loading, defensible within DOE-anchored r_h envelope [$\$1.11, \1.62].
- Rigorous value $\$1.02/\text{gal}$ is within the r_h uncertainty range — the headline result is not knife-edge.
- This is the kind of disclosure R&R Round 2 R2 C5 wanted on the record.

Main Result 2: CfD-with-Floor Dominance

Theorem: CfD-with-Floor Weak Pareto Dominance

Theorem (Theorem (CfD-with-floor dominance)). Under A1–A13, in the regime $\alpha \geq \pi_H$, the CfD-with-floor instrument

$$\sigma^*(\omega) = \max\{\bar{p} - p_{\text{SAF}}(\omega), 0\} + s_0^*$$

with strike $K^* = \bar{p}$ (producer break-even price) and floor s_0^* pinned by the NPV constraint:

- **Weakly Pareto dominates** the linear credit $\sigma^{\text{lin}} = \sigma_{\text{prod}}$ at MCPF = 1.
- **Strictly Pareto dominates** at $\mu_W > 1$ via the MCPF transfer-differential channel.

Important: the result is *weak* at MCPF = 1

Producer-surplus and fiscal-transfer contributions cancel under common-measure planner welfare evaluation; **strict** dominance comes from the MCPF channel only at $\mu_W > 1$.

Proof Technique: Borch–Wilson under Restricted $\mathcal{S}$

Two lemmas + one mechanism proposition (proof outlines in backup):

Lemma B.0 (Distorted-probability representation)

In the regime $\alpha \geq \pi_H$, the 2-state mean-CVaR functional collapses to a distorted-expected-utility functional with state weights that overweight L .

This is the technical key that makes Borch–Wilson directly applicable.

Lemma B.1 (Borch–Wilson FOC via LP-dual)

The welfare-optimal transfer $\sigma(\omega) \in \mathcal{S}$ satisfies an equalized-marginal-utility-ratio FOC derived from the fiscal-cost minimization LP-dual of the planner's σ -form choice problem.

Proposition (H-M departure mechanism, H-M departure proposition)

Three structural primitives drive dominance:

- (i) Restricted $\mathcal{S}$ (A11),
- (ii) Mean-CVaR with $\lambda_{\text{prod}} > \lambda_{\text{planner}}$ (i.e., $\lambda_{\text{prod}} - \lambda_{\text{planner}} > 0$),
- (iii) Project-finance hurdle $r_h > r_{\text{RN}}$.

At least 2 of 3 are necessary for CfD dominance; 45Z satisfies all 3.

Why CfD-with-Floor Wins: The Mechanism

Two simultaneous efficiency gains in Regime A (NPV-binding)

- 1 **Borch–Wilson risk-sharing.** CfD concentrates payments in the low-price tail where the producer's marginal utility of income is highest. Equalizes the marginal-utility ratio between regulator and producer.
- 2 **Fiscal-cost minimization.** CfD pays in low-price states only (plus floor); its expected fiscal outlay $\mathbb{E}[T_{\text{CfD}}] < \mathbb{E}[T_{\text{lin}}]$ at the same producer participation NPV.

Why linear loses

- Linear σ_0 pays full credit even in H -state when producer's marginal utility is low *and* when SAF is profitable on operating margin alone.
- Under A11 restricted S , the planner cannot escape via fully state-contingent instruments — but *within* S , CfD is best.
- Without A11, H-M Theorem 7 logic would recover linear optimality (CARA-Normal + unrestricted contract space).

Corollary: MCPF Differential and Actual Welfare Gain

Corollary (MCPF corollary). Expected *transfer-cost* differential between linear σ and CfD-with-floor (planner's unconditional weighting):

$$\underbrace{\mathbb{E}[T_{\text{lin}}] - \mathbb{E}[T_{\text{CfD}}]}_{\text{transfer-cost differential, not welfare gain}} = \lambda_{\text{prod}} \cdot \pi_H \cdot (\bar{p} - a_L) \cdot Q^* = \$0.292/\text{gal}$$

Actual welfare gain in Regime A (NPV-binding), via the MCPF channel at $\mu_W > 1$:

$$\Delta W_{\text{CfD vs lin}} = (\mu_W - 1) \cdot \$0.292/\text{gal} + O((\mu_W - 1)^2)$$

At Dahlby empirical US-federal MCPF range $\mu_W \in [1.10, 1.30]$

- Welfare gain $\in [\$0.029, \$0.088]/\text{gal}$.
- At SAF Grand Challenge 3 B gal/yr: \$87M–\$264M/yr.
- $\approx 2\%$ – 6% of post-H.R. 1 45Z fiscal outlay (\$4.08B/yr).

The \$7/gal “first-best fiscal-slack diagnostic” is a Pigouvian counterfactual ceiling (§ Theorem Step 4b), *not* an achievable welfare gain within $\mathcal{S}$.

Robustness

Robustness 1: MCPF $\mu_W > 1$ (Dahlby range)

Following Cullen 2013 convention, robustness to μ_W

- A12 sets $\mu_W = 1.20$ as headline (Dahlby midpoint); $\mu_W = 1$ is the welfare-neutral reference case. Range: $\mu_W \in [1.10, 1.30]$.
- Welfare gain $\Delta W = (\mu_W - 1) \cdot \$0.292/\text{gal}$:
 - $\mu_W = 1.10$: $\Delta W = \$0.029/\text{gal} \Rightarrow \$87\text{M}/\text{yr}$ at 3 B gal.
 - $\mu_W = 1.20$: $\Delta W = \$0.058/\text{gal} \Rightarrow \$175\text{M}/\text{yr}$.
 - $\mu_W = 1.30$: $\Delta W = \$0.088/\text{gal} \Rightarrow \$264\text{M}/\text{yr}$.
- **CO₂ cost-effectiveness**: at GREET-implied abatement of ≈ 10.4 kg CO₂e/gal, welfare gain $\approx \$2.8$ – $\$8.5/\text{ton CO}_2$ ($\approx \$5.6$ at the $\mu_W = 1.20$ headline).
- EPA central SCC = $\$190/\text{ton}$ (2020\$) $\Rightarrow$ gain ≈ 1.5 – 4.5% of SCC.
- *Modest in magnitude but sign-determinate across the empirical MCPF range.*

Robustness 2: Private Hurdle IRR (§6.5, editor-flagged)

If r_h is private (relaxes A10): how much does the 87.8% share shrink?

Anonymous σ -form must clear worst-case hurdle:

$$\sigma_{\text{paid}}(\sigma_r) \approx \sigma_{\text{prod}}(\bar{r}_h) + \frac{\partial \sigma_{\text{prod}}}{\partial r_h} \cdot \sqrt{3} \sigma_r,$$

leakage rate $\sqrt{3} \cdot \frac{\partial \sigma_{\text{prod}}}{\partial r_h} = \sqrt{3} \cdot 11.96 = 20.7/\text{gal}$ per unit σ_r .

σ_r	r_h range	C ₂ effective	C2 share
0 (public)	{0.125}	\$0.739	87.8%
0.014 (empirical)	[0.100, 0.150]	\$0.443	81.1%
0.021 (threshold)	[0.089, 0.161]	\$0.304	75.0%
0.025 (stress)	[0.082, 0.168]	\$0.221	68.2%

- 75% threshold reached at $\sigma_r = 0.021$ — **50% wider than empirical DOE range**.
- Variance of private r_h would have to inflate 2.25× before the 12.2%/87.8% attribution qualitatively weakens.
- Heterogeneous-investor screening deferred to Paper 3.

Robustness 3: Extension to 45V / 45Y / 45Q

Credit	Unit	κ_{CAPEX}	r_h	T	$\lambda_{\text{prod}} - \lambda_{\text{planner}}$
45Z (SAF)	\$/gal	11.0	0.125	10	+0.10
45V (H ₂)	\$/kg	≈ 5	≈ 0.12	10	$\approx +0.08$
45Y (electricity PTC)	\$/MWh	≈ 0.5	≈ 0.08	20–30	$\approx +0.05$
45Q (CCS)	\$/ton	≈ 50	≈ 0.10	12	$\approx +0.07$

- All four credits satisfy the H-M departure primitives (i)–(iii) of H-M departure proposition.
- CfD-with-floor dominance result applies separately to each credit.
- Wedge magnitude scales in κ_{CAPEX} .
- Financial-frictions component scales in $(r_h - r_{\text{RN}})/r_{\text{RN}}$.
- Risk-aversion component scales in $(\lambda_{\text{prod}} - \lambda_{\text{planner}})\pi_H(a_H - a_L)$.
- 45Y (long T) and 45Q (high κ_{CAPEX}) expected to have largest CfD wedges.

Policy Product

From Theorem to CFR Comment Letter

The Q3 2026 CFR rulemaking is asking 3 concrete questions

- 1 Is \$1.00/gal (post-H.R. 1) enough?
- 2 Is the linear per-gallon form the right form?
- 3 How should DOE Title XVII interact with 45Z?

This paper answers all three

- **Q1:** No — the \$1.00 base delivers \$0.74–\$0.83/gal CI-scaled at FT-FR, a **\$0.53–\$0.62/gal gap** below $\sigma_{\text{prod}} = \$1.36/\text{gal}$.
- **Q2:** No — CfD-with-floor strictly dominates at $\mu_W > 1$ (welfare gain \$87M–\$264M/yr).
- **Q3:** **46% fiscal substitution rate:** Title XVII compression of r_h from 12.5% to 7% lowers σ_{prod} from \$1.36 to \$0.73/gal.

The 46% Title XVII Fiscal Substitution

Mechanism: DOE Title XVII compresses producer hurdle r_h

A federal Title XVII loan guarantee (sovereign credit enhancement) at the project-finance level reduces r_h from 12.5% to $\approx 7\%$ (typical guaranteed-project WACC):

$$r_h = 0.125 \rightarrow r_h^{\text{compressed}} = 0.07$$

$$\sigma_{\text{prod}} = \$1.36/\text{gal} \rightarrow \sigma_{\text{prod}}^{\text{compressed}} = \$0.73/\text{gal}$$

Substitution: \$0.63/gal of 45Z linear-credit outlay redirectable onto Title XVII contingent liability.

- At 3 B gal/yr SAF Grand Challenge scale: **\$1.88B/yr gross substitution.**
- Net of fees + default provisions: **\$1.70B/yr net substitution.**
- Federal balance sheet absorbs residual at Title XVII historical $\approx 2\%$ default rate.
- **Fiscal substitution rate** (gross): $\approx 46\%$ of 45Z outlay at FT-FR baseline.
- *Pathway-conditional* substitution: higher for PtL, FT (CAPEX-heavy); lower for ATJ-corn.

Quantitative Policy Product Summary

Component	Mechanism	Magnitude
Form switch (CfD-with-floor)	MCPF transfer-differential	
at $\mu_W = 1.10$	$(\mu_W - 1) \cdot \$0.292/\text{gal}$	\$87M/yr
at $\mu_W = 1.30$	$(\mu_W - 1) \cdot \$0.292/\text{gal}$	\$264M/yr
CO ₂ cost-effectiveness	$\Delta W/10.4 \text{ kg/gal}$	$\approx \$3\text{--}\$9/\text{ton}$ ($\approx 2\text{--}5\%$ of \$190 SCC)
Title XVII r_h compression	$r_h : 12.5\% \rightarrow 7\%$	
Per-gallon substitution	$\sigma_{\text{prod}} : \$1.36 \rightarrow \0.73	\$0.63/gal
Gross substitution (3 B gal)	46% of 45Z outlay	\$1.88B/yr
Net substitution	after fees + default provision	\$1.70B/yr
H.R. 1 post-2025 level	\$0.74–\$0.83 CI-scaled vs. $\sigma_{\text{prod}} = \$1.36$	–\$0.53–\$0.62/gal gap

Modest welfare gain from form switch; large fiscal substitution from r_h compression. The two are complements, not substitutes.

§7.3 Rulemaking Recommendation: Four Parts

Treasury+DOE joint proposal for Q3 2026 CFR cycle

- 1 **Title XVII tenor:** 12 years = 10-yr project life + 2-yr construction bridge. Aligns with DOE LPO norms (IRS Asset Class 49.5, Notice 2024-49).
- 2 **Fee structure:** 1.5% origination + 0.5%/yr annual fee = 7.5% lifecycle cost on 12-yr tenor (Section 1703 precedent).
- 3 **Statutory amendment:** new IRC §45Z(g) “Contract-for-Difference Election” — IRS+DOE may elect CI-tier-conditional CfD-with-floor administration of 45Z in lieu of linear payment. Election irrevocable at Stage-1 commissioning (A9 form-commitment). Triggers Title XVII eligibility at (1)–(2).
- 4 **Predicted net fiscal substitution:** \$1.70B/yr onto Title XVII contingent liability at 3 B gal/yr FT-FR baseline. Net 45Z outlay drops from \$4.08B/yr to \$2.19B/yr (46% reduction).

This is a comment-letter-ready proposal — not just a theorem.

The Economic Story in One Frame

What is 45Z actually paying for?

- Not (mostly) risk aversion: \$0.10/gal of the \$0.84 wedge at $T = 10$.
- Not (mostly) rent: producer earns hurdle return only.
- **Project-finance hurdle**: \$0.74/gal — the public cost of overcoming the financial-market failure that keeps pioneer plants from clearing the private hurdle at the social cost of capital.

Why the form matters

- Linear σ pays even when SAF is profitable on operating margin.
- CfD-with-floor pays only in low-price tail — aligned with Borch–Wilson risk-sharing and with MCPF fiscal efficiency.
- At $\mu_W > 1$, the welfare gap is strictly positive.

Conclusion

Three Contributions

- 1 **Empirical decomposition.** Closed-form two-component wedge: 12.2% risk aversion + 87.8% financial frictions at FT-FR, $T = 10$. Attribution shifts with T ; level is the policy-robust object.
- 2 **Form theorem.** CfD-with-floor weakly Pareto dominates the linear credit at $MCPF = 1$; strictly dominates at $\mu_W > 1$ via the MCPF channel. Holds under (i) restricted $\mathcal{S}$ (A11), (ii) mean-CVaR with $\lambda_{\text{prod}} > \lambda_{\text{planner}}$, (iii) project-finance $r_h > r_{\text{RN}}$. All three satisfied for 45Z.
- 3 **Policy product.** \$87M–\$264M/yr welfare gain at 3 B gal/yr, \$3–\$9/ton CO₂ abated ($\approx 2\text{--}5\%$ of EPA \$190 SCC), 46% Title XVII fiscal substitution rate, IRC §45Z(g) “CfD Election” statutory amendment proposal.

Limitations

- 1 **Single representative producer.** Heterogeneity in $(\lambda_{\text{prod}}, r_h)$ and screening implications deferred to Paper 3 (direct-mechanism design under private hurdle rates).
- 2 **A10 public info assumption.** §6.5 quantitative bound: $\sigma_r = 0.021$ threshold (50% wider than empirical DOE range); robust within $2.25\times$ variance inflation. Heterogeneous-investor screening = Paper 3.
- 3 **CI heterogeneity.** \$1.36/gal is FT-FR pathway-specific (§6.7). 4-pathway calibration shows full [\$0.54, \$6.35] range across SAF technologies.
- 4 $O(\lambda^2)$ **residual.** Full numerical ΔW at SE-US 248/34 MILP instance reserved for follow-up.
- 5 **Pre-45Z TEA data.** Posterior $(\lambda_{\text{prod}}, r_h)$ will update as post-IRA commercial-deployment data accumulates.

Future Work

- 1 **Paper 3.** Heterogeneous-investor screening under private $(r_h, \lambda_{\text{prod}})$. Revelation Principle + direct mechanism design. Question: optimal menu of σ -forms across pioneer-plant types.
- 2 **Numerical companion.** Full SP-MILP at 248/34 with $O(\lambda^2)$ second-order welfare terms and binary facility-siting dynamics.
- 3 **FOAK $\rightarrow$ NOAK learning.** Endogenize $(r_h, \kappa_{\text{CAPEX}})$ along the pioneer-plant learning curve (Lazard (2024)). Question: how does σ_{prod} evolve as learning compresses both CAPEX and equity risk premium?
- 4 **Endogenous MCPF.** Move from Dahlby exogenous μ_W to explicit Treasury budget constraint with Title XVII contingent-liability accounting.

Closing

A production credit calibrated to the project-finance hurdle rate is not a subsidy to producers in the ordinary sense; it is the public cost of overcoming the financial-market failure that prevents pioneer plants from clearing the private investment hurdle at the social cost of capital.

Thank you.

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Backup: λ_{prod} Identification (Revealed Preference)

λ_{prod} is **not** an external estimate — it is recovered from the observed 45Z band via the NPV-binding identity, conditional on r_h .

- 1 Fix the project-finance primitives ($r_h = 0.125$, κ_{CAPEX} , u , τ , T) and price primitives (π_H , $a_H - a_L$, $\bar{p} - \bar{c}_{op}$); Farooq (2025) anchors the TEA side, not the CVaR weight.
- 2 Invert $\sigma_{\text{prod}}(\lambda_{\text{prod}}) = \text{CAPEX term} + \lambda_{\text{prod}} \pi_H(a_H - a_L) - (\bar{p} - \bar{c}_{op})$ at the band endpoints [\$1.30, \$1.45].
- 3 Band image: $\lambda_{\text{prod}} \in [0.54, 0.69]$; midpoint 0.60.
- 4 Honest sensitivities: the r_h envelope moves the implied λ_{prod} by $\approx \pm 0.29$ (straddles $\lambda_{\text{planner}} = 0.50$); \$6418 transfer discount (91¢) shifts it to [0.43, 0.56].

Reading: the band-match is a consistency check, not independent validation — the map returns the band by construction.

Backup: Proposition σ_{prod} Proof Outline

Step 1. Stage 2 best response (Lemma Stage 2 Lemma): $Q^*(\sigma; \omega) = (a(\omega) + \sigma - c)/(b + k)$.
Strictly concave in Q ($\partial^2 \pi^{\text{op}} / \partial Q^2 = -(b + k) < 0$).

Step 2. Stage 2 operating profit at Q^* is linear in σ (key structural property).

Step 3. Stage 1 producer's CVaR expectation at $\alpha \geq \pi_H$:

$$\mathbb{E}_{\text{prod}}[\pi^{\text{op}}] = (1 - \lambda_{\text{prod}}) \cdot \bar{\pi} + \lambda_{\text{prod}} \cdot \pi_L \cdot \pi_L^{\text{op}} / \pi_L$$

Reduces to distorted-EU representation (Lemma B.0).

Step 4. Plug into NPV = 0 at hurdle r_h over horizon T :

$$\text{AF}(r_h, T) \cdot \mathbb{E}_{\text{prod}}[\text{CF}(\sigma_0)] / \bar{Q} - \kappa_{\text{CAPEX}} / u = 0$$

Tax shield absorbed into κ_{CAPEX} (absorption convention, §6.6 robustness).

Step 5. Solve for $\sigma_0 = \sigma_{\text{prod}}$:

$$\sigma_{\text{prod}} = \frac{\kappa_{\text{CAPEX}}}{u \text{AF}(r_h, T)(1 - \tau)} + \lambda_{\text{prod}} \pi_H (a_H - a_L) - (\bar{p} - \bar{c}_{op}).$$

□

Backup: Lemma B.0 (Distorted-Probability Representation)

Statement. In the 2-state regime with $\alpha \geq \pi_H$, the mean-CVaR functional

$$\mathcal{V}_\alpha^\lambda[\pi^{\text{op}}] = (1 - \lambda)\mathbb{E}[\pi^{\text{op}}] + \lambda \text{CVaR}_\alpha[\pi^{\text{op}}]$$

admits a *distorted-probability EU representation*:

$$\mathcal{V}_\alpha^\lambda[\pi^{\text{op}}] = q_H^\lambda \cdot \pi_H^{\text{op}} + q_L^\lambda \cdot \pi_L^{\text{op}},$$

with state weights $q_L^\lambda > \pi_L$ and $q_H^\lambda < \pi_H$.

Why it matters. The distorted-EU form makes *Borch–Wilson directly applicable*: the FOC for Pareto-optimal risk-sharing equalizes the marginal-utility ratio across states under the distorted measure. This is the technical bridge between Rockafellar–Uryasev CVaR and the classical Borch–Wilson construction.

Proof. CVaR at $\alpha \geq \pi_H$ in 2 states: $\text{CVaR}_\alpha[X] = \frac{1}{1-\alpha}[(\pi_L - (1-\alpha))X_L + \dots]$, collapses to X_L when $\alpha \geq \pi_H$.

Linear combination with $\mathbb{E}[\cdot]$ recovers distorted-EU form.

Full details: Lemma B.0 in appendix. $\square$

Backup: Lemma B.1 (Borch–Wilson FOC via LP-Dual)

Setup. Fiscal-cost-minimization LP-dual of the planner's σ -form choice problem:

$$\min_{\sigma \in \mathcal{S}} \mathbb{E}[\mu_W \cdot T(\sigma)] \quad \text{s.t.} \quad \mathbb{E}_{\text{prod}}[\pi^{\text{op}}(\sigma)] \geq \bar{\pi}^{\text{op}} \quad (\lambda^{\text{prod}} \text{ multiplier})$$

$$\text{NPV}(\sigma; r_h, T) \geq 0 \quad (\nu^{\text{NPV}} \text{ multiplier})$$

FOC. At interior solution, the Borch–Wilson condition:

$$\frac{q_{\omega}^{\lambda_{\text{planner}}}}{q_{\omega}^{\lambda_{\text{prod}}}} \cdot \mu_W = \lambda^{\text{prod}} + \nu^{\text{NPV}} \cdot \text{AF}(r_h, T) \quad \forall \omega$$

Equalizes the ratio of distorted-state weights between regulator and producer.

Implication. Optimal $\sigma^*(\omega)$ concentrates payments in states where the ratio $q_{\omega}^{\lambda_{\text{planner}}} / q_{\omega}^{\lambda_{\text{prod}}}$ is largest — the low-price tail. CfD-with-floor is the form in $\mathcal{S}$ that approximates this concentration.

Full details: Lemma B.1 in appendix. $\square$

Backup: Theorem CfD-with-Floor Dominance Proof Outline

Step 1. Lemma B.0 collapses both agents' objectives to distorted-EU.

Step 2. Lemma B.1 LP-dual yields Borch–Wilson FOC.

Step 3. Within $\mathcal{S}$ (4 elements), CfD-with-floor uniquely satisfies the Borch–Wilson FOC at the producer's NPV-binding constraint.

Step 4a. At $\text{MCPF} = 1$: producer surplus and fiscal transfer cancel under common-measure planner evaluation $\Rightarrow$ **weak** dominance.

Step 4b. *First-best fiscal-slack diagnostic* (NOT a real welfare gain): if planner could directly substitute r_{RN} for r_h , the counterfactual ceiling is $\kappa_{\text{CAPEX}}/u \cdot [\text{AF}(r_{\text{RN}}, T) - \text{AF}(r_h, T)]/\text{AF}(r_h, T) = \$7/\text{gal}$. This is a Pigouvian ceiling, *outside* $\mathcal{S}$.

Step 4c. At $\mu_W > 1$: MCPF transfer-differential channel activates.

$\mathbb{E}[T_{\text{lin}}] - \mathbb{E}[T_{\text{CfD}}] = \lambda_{\text{prod}} \pi_H(\bar{p} - a_L)Q^* = \$0.292/\text{gal}$.

Step 5. Welfare gain: $\Delta W = (\mu_W - 1) \cdot \$0.292/\text{gal} + O((\mu_W - 1)^2)$.

Step 6. At Dahlby $\mu_W \in [1.10, 1.30]$: $\Delta W \in [\$0.029, \$0.088]/\text{gal}$. **Strict dominance** at $\mu_W > 1$.

Full details: Theorem (CfD-with-floor dominance) in appendix. $\square$

Backup: Proposition — Mechanism for H-M Departure

Statement (H-M departure proposition). CfD-with-floor dominance over linear requires *at least two of three* primitives:

- 1 Restricted $\mathcal{S}$ (A11): otherwise H-M Theorem 7 logic recovers linear optimality under CARA-Normal.
- 2 Producer mean-CVaR with $\lambda_{\text{prod}} > \lambda_{\text{planner}}$: otherwise risk-aversion premium $C_1 = 0$.
- 3 Project-finance hurdle $r_h > r_{\text{RN}}$: otherwise financial-frictions premium $C_2 = 0$.

Application to 45Z. All three hold at the empirical calibration:

- A11: IRC §45 statutory restriction (per-unit-output).
- $\lambda_{\text{prod}} - \lambda_{\text{planner}} = 0.60 - 0.50 = +0.10$ (TEA-calibrated).
- $r_h - r_{\text{RN}} = 0.125 - 0.03 = 0.095$ (three-source anchored).

Application to 45V/45Y/45Q. All three hold qualitatively $\Rightarrow$ CfD-with-floor dominance extends across the IRA production-credit family.

Full details: H-M departure proposition in appendix. $\square$

Backup: Corollary MCPF Differential Transfer (4-step)

Step 1. Compute expected fiscal transfer under linear:

$$\mathbb{E}[T_{\text{lin}}] = \sigma_{\text{prod}} \cdot Q^* = \$1.36 \cdot Q^*/\text{gal} \quad (\text{constant per gallon})$$

Step 2. Compute expected fiscal transfer under CfD-with-floor:

$$\mathbb{E}[T_{\text{CfD}}] = \pi_L \cdot (\bar{p} - a_L) \cdot Q^* + s_0^* \cdot Q^*$$

with s_0^* pinned by $\text{NPV} = 0$.

Step 3. Transfer differential (at $\lambda_{\text{prod}} = 0.60$, $\pi_H = 0.472$, $\bar{p} - a_L = 1.03$):

$$\mathbb{E}[T_{\text{lin}}] - \mathbb{E}[T_{\text{CfD}}] = \lambda_{\text{prod}} \cdot \pi_H \cdot (\bar{p} - a_L) \cdot Q^* = \$0.292/\text{gal}$$

Step 4. Welfare gain at MCPF $\mu_W > 1$:

$$\Delta W_{\text{CfD vs lin}} = (\mu_W - 1) \cdot \$0.292/\text{gal} + O((\mu_W - 1)^2)$$

Why the linear- σ_{prod} formula contains λ_{prod} but the transfer differential also contains λ_{prod} : the producer's CVaR weighting determines both the NPV-binding threshold and the back-end transfer concentration. Full details: MCPF corollary in appendix. $\square$

Backup: §6.6 Rigorous Depreciation Convention

Carrying τD_t through the annuity factor without first-period absorption:

$$\sigma_{\text{prod}}^{\text{rig}} = \frac{\kappa_{\text{CAPEX}}}{u \text{AF}(r_h, T)(1 - \tau)} \cdot \left[1 - \frac{\text{AF}(r_h, T)\tau}{T} \right] + \lambda_{\text{prod}} \pi_H(a_H - a_L) - (\bar{p} - \bar{c}_{op})$$

Baseline: $\tau = 0.21$, $T = 10$, $\text{AF} = 5.5364$.

$\text{AF} \tau / T = 5.5364 \cdot 0.21 / 10 = 0.1163 \Rightarrow \text{bracket} = 0.884$.

Hurdle-IRR-premium term contracts: $\$2.96 \rightarrow \$2.96 \cdot 0.884 = \$2.62/\text{gal}$.

$$\sigma_{\text{prod}}^{\text{rig}} = 2.62 + 0.62 - 2.22 = \$1.02/\text{gal}$$

Where this lands. Below 45Z band but within DOE-anchored r_h uncertainty range [$\$1.11, \1.62] $\Rightarrow$ not knife-edge. Match holds for absorption convention (Brandao 2025; Lazard 2024; Cullen 2013); fails for rigorous derivation.

Honest disclosure per R&R Round 2 R2 C5.

Backup: Asymmetric-Info Robustness Full Table

σ_r	r_h range (uniform G)	Leakage (\$/gal)	C_2 effective (\$/gal)	C2 share	vs. 75% thresh
0 (public)	{0.125}	\$0.000	\$0.739	87.8%	+12.8pp
0.005	[0.116, 0.134]	\$0.104	\$0.635	86.0%	+11.0pp
0.010	[0.108, 0.142]	\$0.207	\$0.532	83.8%	+8.8pp
0.014 (empirical)	[0.100, 0.150]	\$0.296	\$0.443	81.1%	+6.1pp
0.020	[0.090, 0.160]	\$0.414	\$0.325	75.9%	+0.9pp
0.021 (thresh.)	[0.089, 0.161]	\$0.435	\$0.304	75.0%	0.0pp
0.025	[0.082, 0.168]	\$0.518	\$0.221	68.2%	-6.8pp

Threshold at $\sigma_r = 0.021$ corresponds to ± 3.6 pp uncertainty range on r_h — 50% wider than DOE-anchored empirical range. Variance would have to inflate $(0.021/0.014)^2 = 2.25\times$ before 12.2%/87.8% qualitatively weakens.

Backup: Notation Glossary

Symbol	Meaning
σ_{prod}	Producer financial-equivalence subsidy (\$/gal)
$\sigma_{\text{planner}}^*$	Planner welfare-equivalence subsidy (\$/gal); Paper 1 inheritance
$\lambda_{\text{prod}}, \lambda_{\text{planner}}$	Producer / planner mean-CVaR weights ($\in [0, 1]$)
r_h, r_{RN}	Producer hurdle IRR; planner risk-neutral discount rate
κ_{CAPEX}	CAPEX intensity (\$/gal annual capacity)
$\text{AF}(r_h, T)$	Annuity factor at r_h over horizon T
u	Capacity utilization (= 0.85)
τ	Effective federal corporate tax rate (= 0.21)
μ_W	Marginal cost of public funds (Dahlby $\in [1.10, 1.30]$)
π_H, π_L	State probabilities (= 0.472, 0.528)
a_H, a_L	Demand intercepts (\$7.16, \$4.97/gal)
$p_{\text{SAF}}, c_{\text{op}}^-$	Mean SAF price (\$6.00) / op cost (\$3.78) per gal
$\mathcal{S}$	4-element restricted policy class
C_1, C_2	Wedge components (risk aversion / financial frictions)
$\Delta \text{IRR}_{\text{prem}}$	Per-gallon hurdle-IRR premium (Component 2 closed form)

Backup: Key References (Author-Year)

- Li & Yu (2026), *Energy Economics* — inherited $\sigma_{\text{planner}}^* = \$0.52/\text{gal}$ benchmark, SE-US 248/34 SP-MILP.
- Cullen (2013), *AEJ:EP* — closest precedent on PTC welfare; risk-neutral; defers risk aversion.
- Wu et al. (2025), NBER w34326 — direct predecessor; per-gallon credit is least-cost instrument.
- Borch (1962), *Econometrica* — Pareto-optimal risk sharing.
- Wilson (1968), *Econometrica* — syndicates with heterogeneous beliefs.
- Holmstrom & Milgrom (1987), *Econometrica* — linear contract optimality (failure-mode benchmark).
- Newbery (2023), *Energy Policy* — CfD dominance in renewable electricity.
- Maccheroni, Marinacci, Rustichini (2006), *Econometrica* — variational preference foundation for mean-CVaR.
- Bistline et al. (2023, 2024) — IRA-wide assessments + SCC \$190/ton.
- Brandao (2025), Farooq (2025) — 80-TEA harmonization for $(\lambda_{\text{prod}}, r_h, \kappa_{\text{CAPEX}})$.
- Lazard LCOE+ 2024, NREL ATB 2024 — independent calibration cross-checks for r_h .
- DOE SAF GC (2022), IRS Notice 2024-49, CRS R47950 (2025) — policy primary sources.

Backup: Anticipated Q&A Pointers

- **Q: Why $T = 10$?** IRS Asset Class 49.5 + IRS Notice 2024-49 (15-yr alternative SL); DOE Title XVII tenor norm; NREL FOAK 10-15 yr; IEA Net Zero Aviation 2024; Paper 1 inheritance.
- **Q: Isn't the 12/88 split fragile?** Yes — it is a *point estimate* at $T = 10$, not a robust band. Level of the wedge is robust. Attribution shifts: 4/96 at $T = 5$; 71/29 at $T = 20$.
- **Q: Why $\mu_W = 1$ baseline?** Following Cullen 2013 convention. §6.3 robustness to $\mu_W \in [1.10, 1.30]$ (Dahlby).
- **Q: Why is dominance only weak at $MCPF = 1$?** Producer-surplus and fiscal-transfer cancel under common-measure planner evaluation. Strict at $\mu_W > 1$ via the transfer-differential channel.
- **Q: Where does the \$7/gal in the appendix come from?** First-best fiscal-slack diagnostic (Pigouvian ceiling on direct $r_h \rightarrow r_{RN}$ substitution); *not* an achievable welfare gain inside $\mathcal{S}$.
- **Q: What about asymmetric info?** §6.5: 75% C2-share threshold at $\sigma_r = 0.021$ (50% wider than empirical); variance would have to inflate $2.25\times$ to qualitatively weaken result. Paper 3 = full screening.
- **Q: Why CfD-with-floor and not pure CfD?** Floor pinned by NPV-binding constraint; pure CfD doesn't clear hurdle without sufficient strike pricing.